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IN PROGRESS | EE 31 JUNIOR DESIGN | FALL 2026

Autonomous ESP32 Robot

ESP32 | Arduino | IR LED & photodetector | PCB design | Onshape

PROBLEM

- The robot must follow a colored lane and avoid obstacles without human input.
- It must also receive commands from a partner team's robot.
- Every decision depends on sensor readings it can trust.

SO FAR

- Programmed a seven-state state machine on the ESP32, driven by a pushbutton interrupt or a websocket command.
- Selected the IR LED and photodetector resistor values and subtract ambient light from each reading.
- Developing detection logic that filters noise in the color readings before the lane color is identified.

NEXT

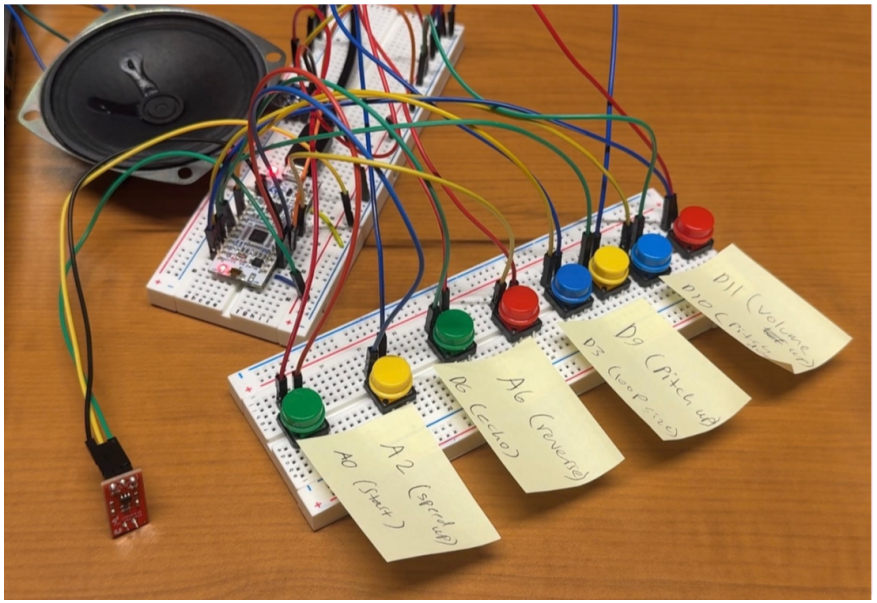
- Designing the robot's custom ESP32 PCB.
- Designing its outer body in Onshape.

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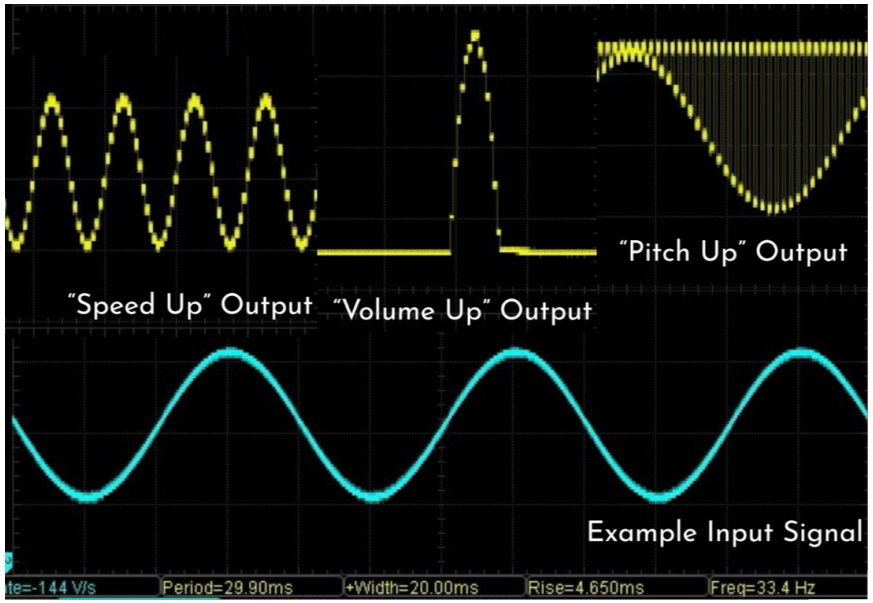
EE 14 EMBEDDED SYSTEMS | JAN. 2026 TO MAY 2026

DJ Board on an STM32

STM32L432KC | C | ADC & DAC | Oscilloscope



The board: microphone, speaker and effect buttons



Oscilloscope captures of the effect outputs

PROBLEM

- A real-time audio effects processor records sound and remixes it through 7 effects.
- Every effect depends on a clean recording.
- The initial playback was marred by a high-pitched hiss and audible clicks.

SOLUTION

- Implemented a timer-driven state machine that records and plays every sample at a constant rate.
- Allowed each microphone reading additional time to complete so that no sample was cut short.
- Held the output at a steady resting level between clips and faded each clip in and out.

RESULT

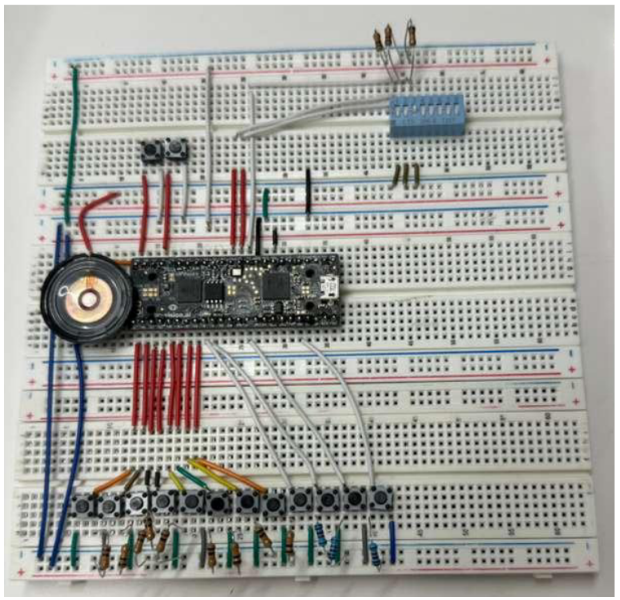
- Oscilloscope measurements confirmed that playback matched the original recording.
- All 7 effects operate within 64KB of on-chip memory.

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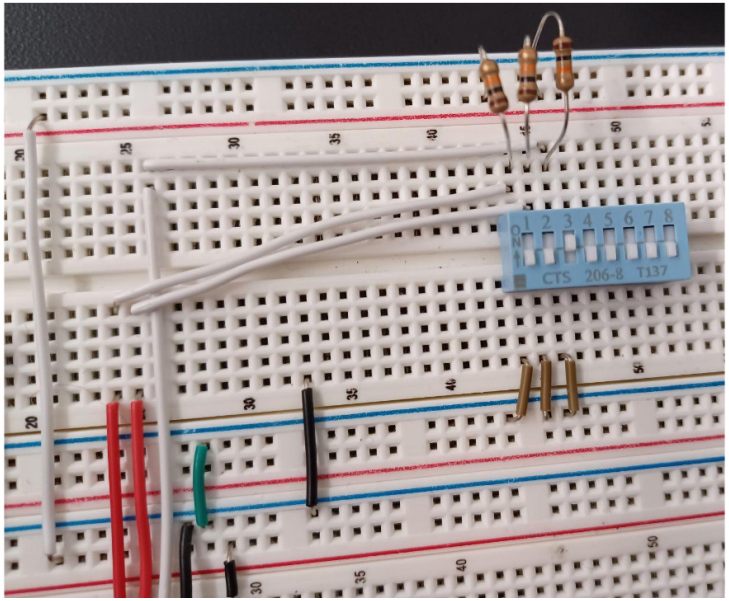
ES 4 DIGITAL LOGIC CIRCUITS | SEPT. 2025 TO DEC. 2025

FPGA Audio Synthesizer

SystemVerilog | FPGA | PWM



The complete board



The switch that selects the octave

PROBLEM

- Each octave doubles a note's frequency.
- The initial design computed every note through division on the chip.
- Division required more than 250% of the FPGA's capacity.

SOLUTION

- Hand-calculated one octave and derived the rest through bit shifts, since dividing by 2 is equivalent to a bit shift.
- Mixed the notes with PWM after testing revealed that chords distorted.

RESULT

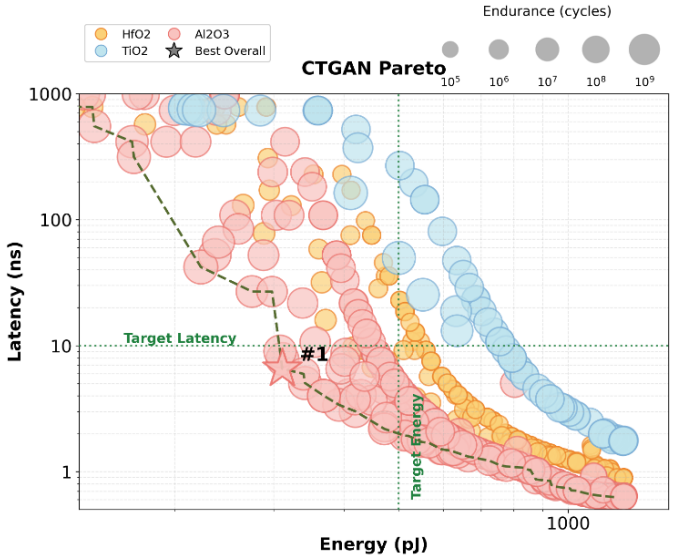
- An 8-octave synthesizer, from 16Hz to 8kHz, for a MIDI keyboard.
- Chords of up to 6 notes play clearly.

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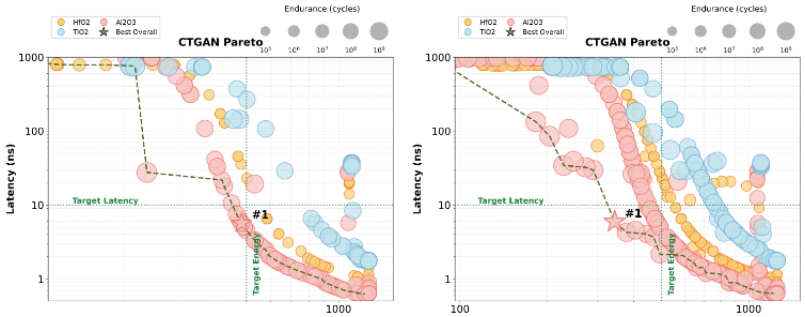
TUFTS EMERGING CIRCUITS AND SYSTEMS LAB | SEPT. 2025 TO APRIL 2026

CTGAN Design Discovery for RRAM

Python | PyTorch | pandas | CUDA | CTGAN | PINN



Final results with 5,000 training samples



The same search with 500 samples (left) and 2,500 (right)

PROBLEM

- A published AI framework designs RRAM memory devices.
- Its model averaged the training data, smoothing away the extreme designs where peak performance lies.

SOLUTION

- Replaced the model with a CTGAN, which explores those extremes more aggressively.
- Scaled the training data tenfold, to 5,000 samples, so that the CTGAN could learn reliably.
- Employed the framework's physics model as a referee to reject impossible designs.

RESULT

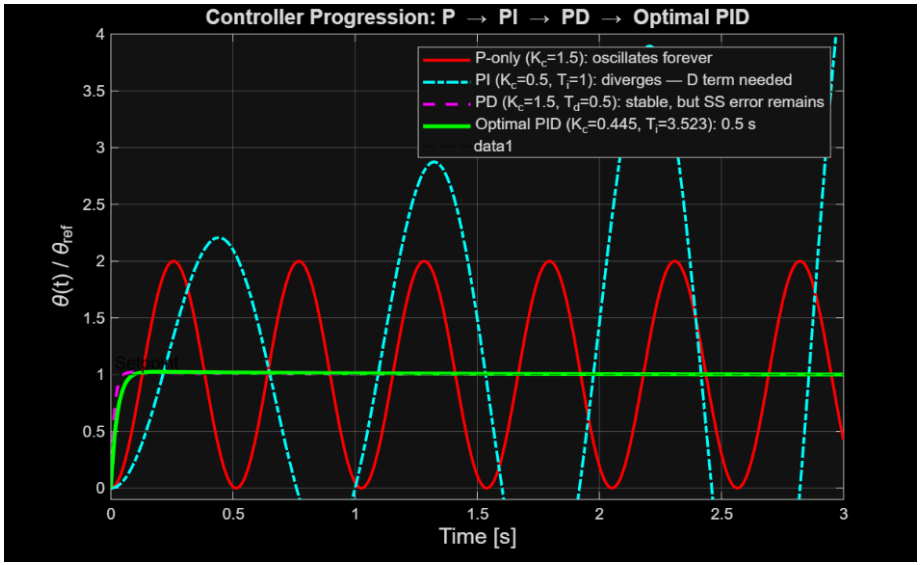
- 25% higher endurance than the published result.
- 11% lower switching energy.
- Slightly slower switching: 6.57ns versus 4.07ns.

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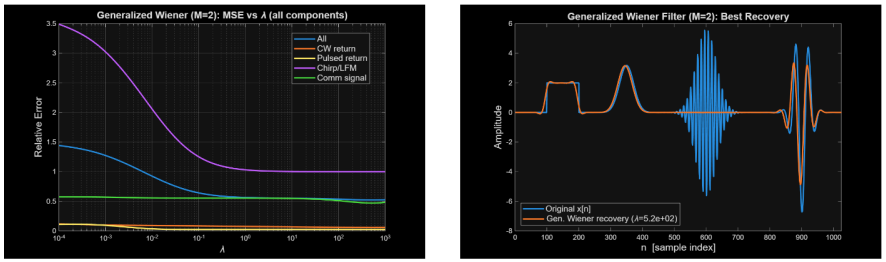
EE 23 LINEAR SYSTEMS | JAN. 2026 TO MAY 2026

UAV Pitch Control and RF Signal Restoration

MATLAB | PID control | Wiener filtering



P, PI, PD and PID controllers on the same UAV model



Wiener filter error (left) and the restored RF signal (right)

PROBLEM

- Stabilizing a UAV in pitch is the first step toward collecting clear RF signals from the air.
- Even a stable UAV shakes slightly and blurs the signals it receives.

SOLUTION

- Demonstrated through Routh-Hurwitz analysis that P and PI controllers cannot stabilize the UAV.
- Searched 15,625 gain combinations for the optimal PID settings.
- Restored a blurred, noisy RF signal with a Wiener filter.

RESULT

- 52ms rise time, 2.6% overshoot and 500ms settling time, within real UAV specifications.
- 77% lower error than a direct inverse (MSE 2.272 to 0.521).